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Python Setup – Windows

System requirement

- Windows 7 or newer
- Python 2.7.*

1. Installing Python 2.7

Follow the below link for Python and pip installation

<https://github.com/BurntSushi/nfldb/wiki/Python-&-pip-Windows-installation>

2. Install the modules: **pypiwin32**, **pyserial**, **pyqtgraph**, **scipy**, **termcolor**, **PyOpenGL** and **matplotlib**.

```
> pip install pyserial pyqtgraph scipy termcolor matplotlib PyOpenGL
```

3. Install the **PyQt4**

Here are Windows wheel packages built by Chris Golke.

<https://www.lfd.uci.edu/~gohlke/pythonlibs/#pyqt4>

In the filenames cp27 means C-python version 2.7, cp35 means python 3.5, etc

Make sure you download the correct whl file (Python version, 32/64 bit), and then use pip to install it e.g:

```
> pip install C:\downloads\PyQt4-4.11.4-cp27-cp27m-win_amd64.whl
```

Should properly install if you are running an x64 build of Python 2.7.

Python Setup – Linux (Ubuntu)

System requirement

- Ubuntu 16.04 or newer
- Python 2.7.*

1. Without installing anything, Ubuntu 16.04 includes Python 2.7. Enter below command to check Python version

```
> python --version
```

If Python 2.7 is not installed then enter below commands to install Python 2.7

```
> sudo apt update
```

```
> sudo apt dist-upgrade
```

```
> sudo apt install python2.7 python-pip
```

2. Update your system software

```
> sudo apt-get update
```

```
> sudo apt-get upgrade
```

3. Install **PIP** by enter below command

```
> sudo apt-get install python-pip
```

4. Once the pip's installation is completed you can verify that it was successful by using below command

```
> pip -V
```

6. Install **PyQt4** and **Tkinter** package.

```
> sudo apt-get install python-qt4
```

```
> sudo apt-get install python-tk
```

7. Install **BlueZ** utility as below

```
> sudo apt-get install bluez
```

8. Enter below command. If it open, then above installation is successfully completed

```
> gatttool -I
```

9. To check the list of nearby Bluetooth devices. Enter below commands.

```
> sudo hcitool lescan
```

10. Install the modules: **pyserial**, **pyqtgraph**, **scipy**, **termcolor**, **PyOpenGL** and **matplotlib**.

```
> sudo pip install pyserial pyqtgraph scipy termcolor matplotlib PyOpenGL
```

Python Setup – Linux (Other)

1. PyCharm Community Edition (<https://www.jetbrains.com/pycharm/download>). PyCharm Community Edition is free and open source.

2. Install below package

- Python 2.7
- PIP
- PyQt4
- Tkinter
- BlueZ

3. Follow Steps number 8 to 10 in previous section [Python Linux\(Ubuntu\)](#)

How to run python scripts

1. Here are some steps to run below Python scripts which are present in Python_scripts folder

- precision_imu.py
- calibrated_data.py
- raw_data.py
- swdr_ble.py

a) To change the parameters like comport, outrate etc, open the script with editor and go to top the page. For example in precision_imu_usb.py

d_graph = 'off line'	# Display graph # 'Real Time' #'Off Line'
log_data = 0	# Save lag data in file write 1 else 0
dlog_file = 'data/bin.txt'	# Data log path directory
out_rate = float(31.25)	# Write Hz rate

```

conn_params = ('COM5', 115200) # Write here serial port or mac address on your
device
conn_type = 'usb'                # write here connectivity type e.g. usb or ble
runtime = float(10)              # Enter time in second

# select acc, gyro, and magno [a0i a1i a2i g0i g1i g2i m0i m1i m2i] for each imu
select_acm = ['11111111', '11111111', '11111111', '11111111', '11111111', '11111111',
'11111111', '11111111', '11111111', '11111111', '11111111', '11111111', '11111111',
'11111111', '11111111', '11111111', '11111111', '11111111', '11111111', '11111111',
'11111111', '11111111', '11111111', '11111111', '11111111', '11111111', '11111111',
'11111111', '11111111']

```

- b) Run the Python scripts as below

```
>python <python script>
```

- c) To find the serial port name, refer the documents

<https://www.inertiaelements.com/oblu/resources/how-to-find-serial-port.pdf>

2. To run Calibration script, follow below steps

- a) Change the **com_port** parameter in the top of **main.py** script with mac address of device. E.g.

```
com_port = '00:A0:50:E4:91:98' # Enter the mac address of device
```

- b) Run main.py on command prompt.

```
>python main.py
```